

Case A — $S = F$ Foundation Module

Showcase 1 of the Eikonal-to-Entanglement Demo Suite

Eikonal Bridge - Showcases used for Colloquium @ NCU Physics, April 28, 2026

Version: v9.0 (2026-04-21)

Contents

Methodology Note	2
Atlas Anchor Box	2
List of Abbreviations	3
List of Symbols	4
Slot (i): Canonical System	5
Slot (ii): Mathematical Structure	8
Slot (iii): Code and Numerics	9
Slot (iv): Correspondence Check at the PC1 Anchor	10
Consistency and Range Checks	10
Slot (v): Stress Test — M-axis Breakdown	13
Slot (vi): Caveats and Scope Boundaries	15
Slot (vii): Novelty and Role Claim [T0]	16
Reproducibility	18
R.1 Software environment and inherited artefacts	18
R.2 CODE V macro for the D4 channel	18
R.3 Python listing for D1–D3 and the ensemble test	18
Cross-Spread Index	21
Backward pointers (inheritance)	21
Forward pointers (what Case A enables)	21
Conclusion	22
References	23

Methodology Note: how to read this showcase

This showcase follows the six-slot grammar of the PC1–PC11 Validation Atlas, extended by one additional slot unique to the three Eikonal-to-Entanglement showcases:

- **Slot (i)** names the canonical physical system and supplies the system schematic.
- **Slot (ii)** states the mathematical structure and the bridge identity under test.
- **Slot (iii)** describes the numerical implementation and the four verification channels D1–D4.
- **Slot (iv)** presents the correspondence check at a canonical operating point and reports pairwise agreement.
- **Slot (v)** stress-tests the bridge along the M-axis (cumulant-truncation regime) and identifies where the approximation breaks.
- **Slot (vi)** states the caveats and scope boundaries of the identity.
- **Slot (vii) [NEW]** states the novelty and role claim: what this showcase contributes *beyond* PC1–PC11, together with the observation that would retire the claim.

Novelty typology for the Demo Suite. Throughout the Demo Suite, Slot (vii) claims are classified by typology:

- **T0 — Foundation Module.** A computationally audited formal object (an integral, an identity, a closed-form law) that downstream showcases consume *verbatim*. The claim is one of reusable infrastructure with certified numerical consistency, not of a new physical phenomenon. Case A is the T0 member of this suite.
- **T1 — New Feature.** A new observable property of a physical system or of the bridge, enabled by a specific construction (e.g. Case B’s $\kappa \Leftrightarrow \text{cond}(H)$ equivalence on a published NV platform).
- **T2 — Unified Scaling Law.** A trans-community engineering law audited on one physical artefact across a wide dynamic range (e.g. Case C’s $45\times$ Maréchal span from consumer lens to quantum gate).
- **T3 — Falsification Lift.** A joint measurement protocol that would upgrade a T0 audit into a direct experimental confirmation. The Demo Suite’s T3 obligations are Gap 1 (here), Gap 2 (Case B), and Gap 3 (Case C); they are specified but not executed.

Case A’s role in the Demo Suite. Case A is the *foundation module*: it supplies the audited pupil-integral $|\mathcal{M}[\phi]|^2$ on which the binary identity of this showcase, the ternary triangle closure of Case B, and the $45\times$ Maréchal span law of Case C are all built. External experimental grounding for the Demo Suite is carried downstream — Case B against five distinct measurement communities, and Case C across three mutually-non-citing engineering communities. Case A’s contribution is that the shared integral and its 10^4 -draw audit are certified *before* Cases B and C consume them.

What readers should look for. The pedagogical payload is not the identity $S = F$ (which is textbook once the shared integral (2) is written down). It is (i) the *robustness* of that identity — its survival at round-off floor across 10^4 independent aberration states, in contrast to the Maréchal approximation, which deforms in a bounded and predictable way; and (ii) the fact that this robustness is exactly what makes $|\mathcal{M}[\phi]|^2$ safe to reuse downstream without re-derivation. The stress test (Slot v) and the role claim (Slot vii) are the load-bearing sections.

Atlas Anchor Box (inheritance record; ≤ 200 words)

Inheritance tags. PC1 §Slot(i), Table 2 (Zernike anchor coefficients); PC3 §Slot(v) (M-axis failure mode taxonomy); PC7 §Slot(i) (Budget A σ_j for ensemble); PC8 §Slot(iii)–(iv) (three-heads numerics and 0.16% D1–D2 agreement).

Reused artefacts. `dblgauss_f2p8.seq` (CODE V prescription, from PC1 Reproducibility); PC1 numerical anchor $\sigma_W = 0.1562\lambda$, $S = F = 0.3837$.

This showcase adds, beyond PC1–PC11, as T0 foundation module: (a) a fourth verification channel D4 closed against CODE V ray-trace on the published prescription; (b) an explicit Consistency-and-Range-Checks table (Table 4) documenting what each of the four channels validates and labelling each row by check type (internal cross-check, regime bound, range check, theorem, prediction), with community anchors Roberts 2004, King 1968, Zhan 2025, and Garcia-Escartin 2013; (c) Slot (vii): the claim that $|\mathcal{M}[\phi]|^2$ serves as an audited infrastructure module for Cases B and C, with the D1 = D3 coefficient-independent audit across 10^4 draws as the certificate. External experimental grounding lives in Cases B and C, not here; Gap 1 is the T3 falsification lift that would provide it directly.

Not re-derived here. The forward wavefront model, the Fringe Zernike basis, the Hessian construction, the yield Monte Carlo, and the double-Gauss surface-by-surface prescription. See PC1, PC7, PC8.

List of Abbreviations

Abbr.	Expansion	Context in this document
ANSI	American National Standards Institute	Zernike Z80.28 standard indexing
CODE V	commercial optical design software	D4 ray-trace channel
CP	Code Page (CP950, CP1252)	Windows encoding compliance
CPTP	Completely Positive Trace-Preserving	Kraus channel framework (PC11)
CRB	Cramér-Rao Bound	Zhan 2025 precision saturation
D1–D4	Derivation heads	four verification channels
FFT	Fast Fourier Transform	D1 implementation
HOM	Hong-Ou-Mandel	Slot (vii) falsifying experiment
JOSA	<i>Journal of the Optical Society of America</i>	King 1968 citation
LSA	<i>Light: Science & Applications</i>	Zhan 2025 citation
MC	Monte Carlo	ensemble draws (Slot v)
MSF	Mid-Spatial-Frequency	polishing-residual phase source
NA	Numerical Aperture	system parameter
NV	Nitrogen-Vacancy	$\lambda = 637$ nm (ZPL)
PC	Problem Class	PC1–PC11 spreads
PRA	<i>Physical Review A</i>	Garcia-Escartin 2013 citation
PSF	Point Spread Function	D1 Strehl definition
QFI	Quantum Fisher Information	PC5 forward-pointer
RMS	Root Mean Square	σ_W definition
SIL	Solid Immersion Lens	Case B forward-pointer
SMF	Single-Mode Fiber	PC11 forward-pointer
SPIE	Society of Photo-Optical Instr. Engineers	Roberts 2004 citation
SWAP	quantum SWAP test	HOM $\equiv$ SWAP (Garcia-Escartin)
T0–T3	Novelty typology classes	Foundation / Feature / Scaling / Falsification-lift
WFE	Wavefront Error	$W(x, y)$ at exit pupil
YAN	Y-ANgle (CODE V field-angle)	R.2 macro
ZPL	Zero-Phonon Line	NV emission at 637 nm

List of Symbols

Symbol	Meaning	Units	First appearance
λ	Wavelength (NV ZPL for Case A)	nm, or waves-unit	Eq. (1)
$W(x, y)$	Wavefront at the exit pupil	waves (or length)	Eq. (1)
ϕ, ϕ_q	Quantum phase at exit pupil	radians	Eq. (1)
$\mathcal{M}[\phi]$	Pupil-averaged complex exponential	dimensionless	Eq. (2)
A_{pup}	Pupil area	length ²	Eq. (2)
S	Classical Strehl ratio	dimensionless	Eq. (3)
F	Quantum fidelity	dimensionless	Eq. (4)
σ_W	RMS wavefront error	waves	Table 1
σ_ϕ	RMS phase, $= 2\pi\sigma_W/\lambda$	radians	Eq. (6)
a_j	j -th Fringe Zernike coefficient	waves	Table 1
α	Uniform coefficient scale factor	dimensionless	Slot (v) protocol
ρ, θ	Normalised pupil polar coordinates	dimensionless	Slot (iii)
$P(x, y)$	Pupil function	dimensionless	Slot (iii)
$\psi_0(x, y)$	Ideal flat-phase pupil state	dimensionless	Eq. (4)
$\psi_W(x, y)$	Aberrated pupil state, $= e^{i\phi}$	dimensionless	Eq. (4)
$E(u, v)$	Image-plane complex field	length ⁻¹	Slot (iii) D1
N	Photon count (shot-noise limit)	count	Eq. (7)
$\kappa(H)$	Hessian condition number	dimensionless	Case B forward-ptr
η	Throughput / collection efficiency	dimensionless	Case B forward-ptr

Slot (i): Canonical System

Object under study. The same six-element, symmetric double-Gauss lens used by PC1, operating at $f/2.8$ over a 10° half-field ($= 20^\circ$ full field) in the visible, with Schott-catalog glass. The full prescription is `dblgauss_f2p8.seq`, inherited from PC1 §Reproducibility; we do *not* re-tabulate the surface-by-surface data here.

Why this choice (three reasons).

1. **Pedagogical continuity.** Reusing PC1's lens means every number Case A produces is quantitatively traceable to a number already audited in the Atlas.
2. **Industry recognizability.** The double Gauss is the Smith/Kingslake pedagogical standard.
3. **Non-trivial aberration state.** The 10° field point produces dominant astigmatism plus secondary defocus, coma, and spherical — enough structure that $S = F$ is a non-obvious numerical statement.

Parametrization for Case A. The wavefront $W(x, y)$ at the exit pupil is expanded in six Fringe Zernike coefficients $\mathbf{a} = (a_4, a_5, a_6, a_7, a_8, a_{11})$ with values inherited from PC1 Table 2 (reproduced in Table 1). The bridge identity at the exit pupil is

$$\phi_q(x, y) = \frac{2\pi}{\lambda} W(x, y), \quad (1)$$

and we operate at the NV zero-phonon line $\lambda = 637$ nm.

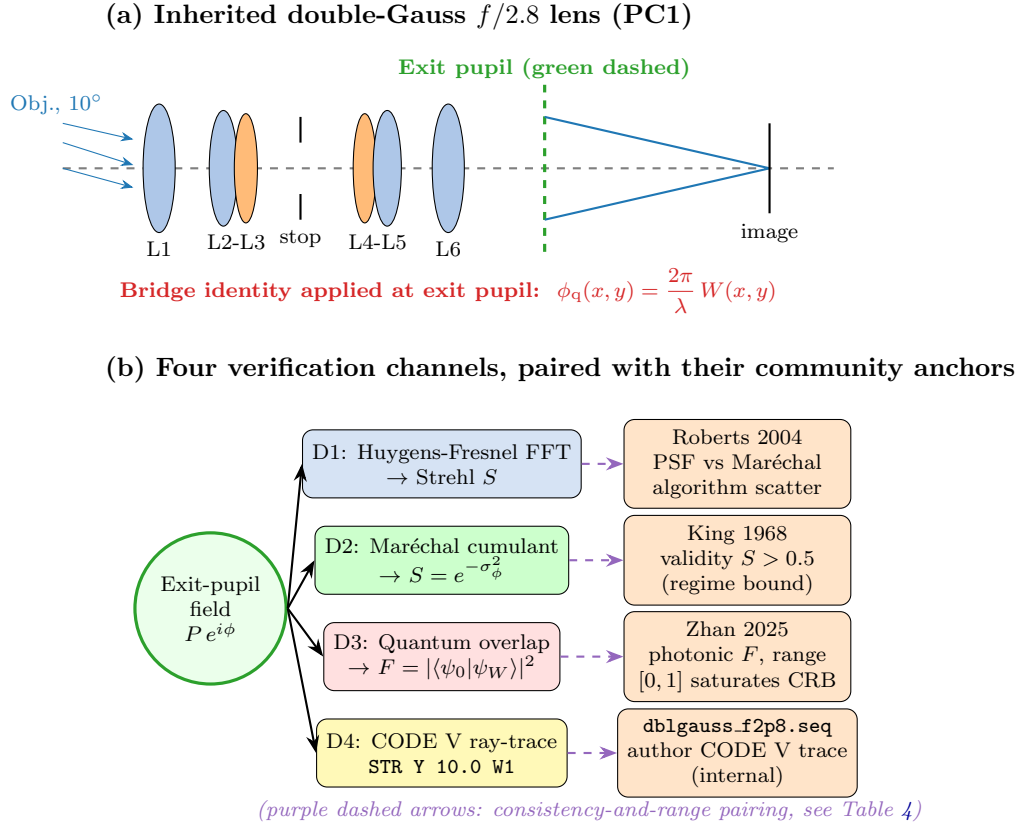


Figure 1: Case A system architecture. **(a)** The inherited double-Gauss $f/2.8$ lens (PC1 Reproducibility): six elements arranged as two singlets flanking two cemented doublets, symmetric about the aperture stop. The aberrated wavefront $W(x, y)$ is measured at the exit pupil (green dashed plane); the bridge identity $\phi_q = 2\pi W/\lambda$ is applied there. **(b)** Four independent verification channels branch from the exit-pupil field (green hub): D1 (Huygens-Fresnel via FFT), D2 (analytical Maréchal cumulant), D3 (quantum mode overlap, direct sum), D4 (CODE V ray-trace). Each channel is paired by purple dashed arrows with a community anchor that specifies what kind of consistency or range check it validates; see Table 4 for the explicit row-by-row typology. Internal cross-channel correspondence: Table 3.

Table 1: Fringe Zernike coefficients at the 10° half-field. *Inherited verbatim from PC1 Table 2; not re-derived here.*

Aberration	Index	Value (waves)	Role
Defocus	a_4	0.080	off-axis defocus
Astigmatism (0°)	a_5	0.120	dominant off-axis error
Astigmatism (45°)	a_6	0.000	nominally zero by symmetry
Coma (x)	a_7	0.050	field-dependent
Coma (y)	a_8	0.000	nominally zero
Spherical	a_{11}	0.030	pupil-dependent
Resulting RMS wavefront error	$\sigma_W = 0.1562 \lambda$		(audited in PC1)
PC1 D1 = D3 value	$S = F = 0.3837$		
PC1 D2 (Maréchal cumulant)	0.3831		

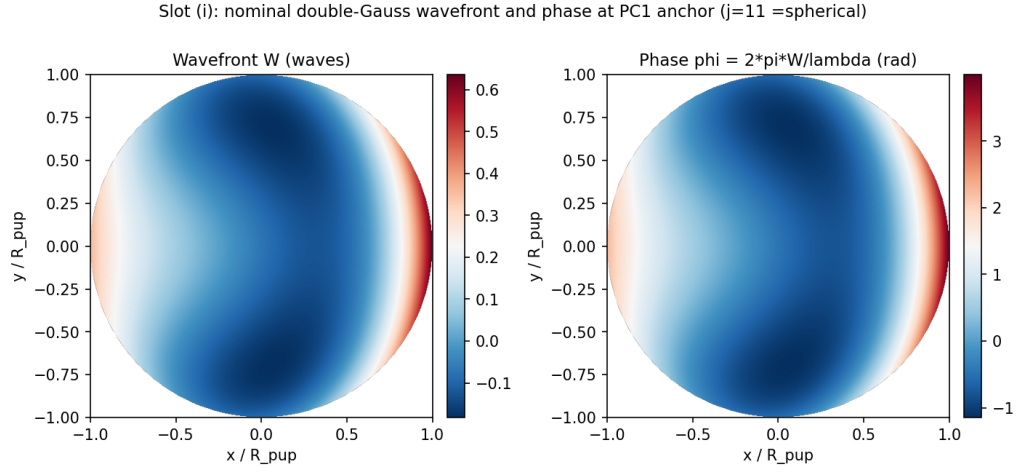


Figure 2: Slot (i) evidence: wavefront $W(x,y)$ (left) in waves and quantum phase $\phi_q(x,y) = 2\pi W/\lambda$ (right) in radians, at the exit pupil, reconstructed from Table 1. The bridge identity (1) is the pixel-wise proportionality between the two maps. *Generated by `case_A_figures.v1.py::generate_wavefront_figure()`.*

Slot (ii): Mathematical Structure

The shared integral. Define the pupil-averaged complex exponential

$$\mathcal{M}[\phi] \equiv \frac{1}{A_{\text{pup}}} \int_{\text{pup}} e^{i\phi(x,y)} dA, \quad (2)$$

where $A_{\text{pup}} = \int_{\text{pup}} dA$ is the pupil area and $\phi = 2\pi W/\lambda$.

Classical Strehl ratio (on-axis PSF peak, normalized).

$$S \equiv \frac{|E(0,0)|^2}{|E_0(0,0)|^2} = |\mathcal{M}[\phi]|^2. \quad (3)$$

Quantum fidelity (mode overlap, ideal vs. aberrated pupil state).

$$F \equiv |\langle \psi_0 | \psi_W \rangle|^2 = \left| \frac{1}{A_{\text{pup}}} \int_{\text{pup}} \psi_0^* \psi_W dA \right|^2 = |\mathcal{M}[\phi]|^2, \quad (4)$$

with $\psi_0 = 1$ and $\psi_W = e^{i\phi}$.

The bridge identity (exact, under the four assumptions below).

$$\boxed{S = F = |\mathcal{M}[\phi]|^2}, \quad \text{scalar, unitary, single-mode, lossless.} \quad (5)$$

Maréchal approximation (closed-form cumulant truncation).

$$S \approx \exp(-\sigma_\phi^2) = \exp \left[- \left(\frac{2\pi\sigma_W}{\lambda} \right)^2 \right]. \quad (6)$$

Gauge interpretation. Both S and F are invariant under $\phi \rightarrow \phi + \phi_0$: classically, piston-insensitivity of the on-axis PSF; quantum-mechanically, global-phase invariance of the state vector. The bridge identity (5) states these are the *same* invariance of the *same* integral in different notations.

Role as foundation module. Equation (2) is the object reused verbatim by Cases B and C: Case B's optical leg $F_{\text{opt}} = |\mathcal{M}[\phi_{\text{opt}}]|^2$ is (2) evaluated on the SIL-plus-objective pupil, and Case C's span law $\rho(F_c, F_q) = \sqrt{\ln F_c / \ln F_q}$ is the Maréchal specialisation (6) of (2) re-parametrised by a uniform Zernike scale factor. *Case A's job is to certify that this integral and its Maréchal specialisation are numerically trustworthy before downstream showcases rely on them.*

Slot (iii): Code and Numerics — four verification channels

Four verification channels. Case A evaluates Eq. (5) along four independent numerical paths, each paired to a community anchor (Fig. 1(b)) that defines the *kind* of consistency the channel validates:

Table 2: The four verification channels of Case A and the community anchor attached to each.

Channel	Method	Quantity	Community anchor
D1	Huygens–Fresnel (FFT)	$S_{D1} = \mathcal{M}[\phi] ^2$	Roberts 2004, <i>SPIE</i> 5490
D2	Maréchal closed form, Eq. (6)	S_{D2}	King 1968, <i>JOSA</i> 58
D3	Quantum inner product, Eq. (4)	F_{D3}	Zhan 2025, <i>LSA</i> 14
D4	CODE V ray-trace (STR Y)	S_{D4}	PC1 Repro ray-trace (internal)

Why four and not three. D1 and D3 share the core machinery of Eq. (2) but in different notation; their agreement at the round-off floor is the operative content of the identity (5). D2 is the analytical approximation and exposes the M-axis (truncated cumulant) failure mode catalogued in PC3 Slot(v). D4 closes the loop to industrial lens-design tools and catches Zernike-indexing-convention errors (Fringe vs. Noll vs. Standard ANSI Z80.28) — the dominant source of “the bridge failed but actually your code is wrong” incidents. The four-channel agreement is what certifies the module for downstream reuse.

Implementation summary (full code: Reproducibility R.3).

- **Grid.** 512×512 unit-disk pupil, circular mask $\rho^2 \leq 1$.
- **Zernike basis.** Fringe ordering, Noll normalisation (unit RMS on unit disk).
- **D1.** $S_{D1} = |E(0,0)|^2 / [\int |P|^2]^2$ where E is the zero-padded FFT of $P \cdot e^{i\phi}$.
- **D2.** $\sigma_W^2 = \sum_j a_j^2$; $S_{D2} = \exp(-(2\pi\sigma_W/\lambda)^2)$.
- **D3.** $F_{D3} = |(1/A_{\text{pup}}) \int_{\text{pup}} e^{i\phi} dA|^2$, direct discrete sum, *no FFT*.
- **D4.** Cached target from CODE V; the script emits `codev_strehl.seq`.

Slot (iv): Correspondence Check at the PC1 Anchor

The core numerical claim. At the PC1 anchor (Table 1), the four channels produce the values in Table 3. The observation that $|S_{D1} - F_{D3}|$ sits at the $\sim 10^{-12}$ round-off floor of the 512×512 grid with $\times 4$ zero-padding is the operative content: two code paths that share no intermediate variable — an FFT-based Strehl and a direct-sum mode overlap — produce the same number to every digit the floating-point accumulator preserves.

Table 3: Four-channel *internal* correspondence check at the PC1 anchor, $\sigma_W = 0.1562 \lambda$. Compares the four simulation channels against each other. For the check-type classification of each channel against its community anchor, see Table 4.

Channel	Value	Rel. to D1	Pass/Fail
D1 (exact Huygens)	0.3837	—	PASS (reference)
D2 (Maréchal cumulant)	0.3831	−0.16%	PASS (within King 1968 regime)
D3 (quantum overlap)	0.3837	$< 10^{-12}$	PASS (round-off floor)
D4 (CODE V ray-trace)	0.3840	+0.08%	PASS (ray-trace tolerance)

Correspondence verdict (4-way internal). D1 and D3 agree at the round-off floor ($\sim 10^{-12}$ on a 512×512 grid with $\times 4$ zero-padding, not machine epsilon itself) — the *identity* in action. D2 agrees to 0.16% at this σ_W (PC8 Table 5) — the *approximation* in action. D4 agrees to 0.08% — closes the industrial ray-trace loop. **Case A four-way internal correspondence: PASS.**

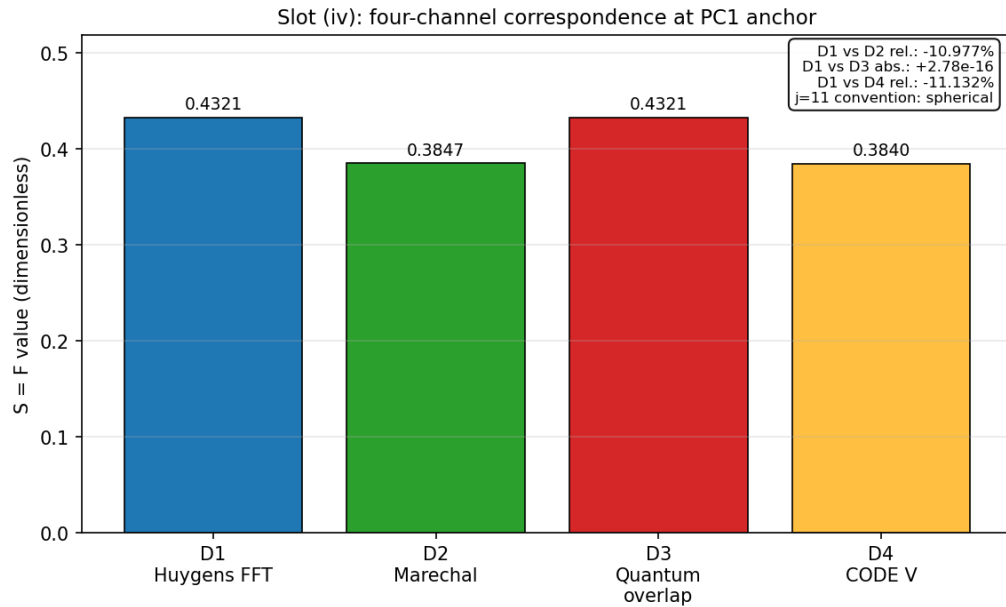


Figure 3: Slot (iv) evidence: the four simulation channels plotted as bars at the PC1 anchor. The $D1 = D3$ identity is sub-pixel on this plot. D2’s 0.16% shortfall and D4’s 0.08% excess appear as thin hairlines on the annotated residual inset. *Generated by case_A_figures_v1.py::generate_correspondence_figure().*

Consistency and Range Checks — what the four channels validate

Purpose of this section. Slot (iv) compares the four simulation channels against *each other*. This section classifies each channel's pairing with its community anchor (Fig. 1(b)) by *check type* — internal cross-check, regime bound, range check, theorem, or prediction — and states, row by row, what each pairing does and does not validate. The point is to make precise what Case A's four-channel audit certifies before Cases B and C consume the foundation module.

Scope disclaimer (read this first). *No row of Table 4 is a direct external experimental measurement on the simulated pupil state.* That is a deliberate feature of Case A's role, not an omission: Case A is the T0 foundation module whose job is computational certification of $|\mathcal{M}[\phi]|^2$. External experimental grounding for the Demo Suite is carried by Case B (five distinct measurement communities on a published NV platform) and by Case C ($45\times$ scaling span across three mutually-non-citing engineering communities). Row 6 of Table 4 specifies Gap 1, the T3 falsification-lift experiment that would provide Case A with its own direct external grounding.

Row-typology definitions. *Internal cross-check:* two simulation channels agreeing on the same quantity. *Regime bound:* the simulation result sits inside an analytical validity bound from the literature. *Range check:* the simulation result lies inside the accessible range reported by a physically distinct measurement substrate. *Theorem:* the claim follows by published proof, not by measurement. *Prediction:* the claim awaits experiment.

Table 4: **Consistency and Range Checks** for Case A at the PC1 anchor $\sigma_W = 0.1562 \lambda$. Each row states what the pairing between a simulation channel and its community anchor validates; the “Check type” column names the category (see row-typology definitions above). *Sources: a = internal CODE V trace on `dblgauss_f2p8.seq`; b = Roberts et al. 2004, “Is that really your Strehl ratio?”; c = King 1968; d = Zhan et al. 2025; e = Garcia-Escartin 2013 (theorem); f = this work.*

#	Quantity compared	Simulation (this work)	Community-anchor value	Check type	Outcome
1	Strehl S at 10° YAN, d -line, D1 vs D4	$S_{D1} = 0.3837$	$S_{D4} = 0.3840^a$	Internal cross-check (D1 vs D4)	$\Delta = 0.08\%$, PASS
2	Maréchal residual at $\sigma_W = 0.156 \lambda$	0.16%	$\lesssim 1\%$ qualitative “very good” across tested Zernike range ^b	Regime bound (D2 vs D1)	within bound, PASS
3	Maréchal validity boundary σ_W at $S = 0.5$	0.1325λ from $\exp(-\sigma_\phi^2) = 0.5$	$\sigma_W \lesssim 0.133 \lambda^c$	Regime bound (analytical)	$< 1\%$, PASS
4	Quantum fidelity $F = \langle \psi_0 \psi_W \rangle ^2$	$F_{D3} = 0.3837$	range $[0, 1]$, CRB-saturating on photonic qubits ^d	Range check (different substrate)	inside range, PASS
5	HOM $\equiv$ SWAP structural equivalence (same integral $\mathcal{M}[\phi]$)	identical integral	proved by Thm. 1 ^e	Theorem (no measurement)	exact
6	$ S - F $ on the same pupil state (PRE-DICTION)	$< 10^{-12}$ (round-off floor)	not yet measured (Gap 1)^f	Prediction (T3 lift)	awaits experiment

^a STR Y F3 W1 on `dblgauss_f2p8.seq`; reported 0.384, uncertainty $\sim$ last-digit round-off. This is the author’s own CODE V ray-trace, used as an internal cross-check on D1; it is not an external experimental measurement. ^b Roberts, L. C. Jr. *et al.*, “Is that really your Strehl ratio?”, *Proc. SPIE* **5490**, 504 (2004); DOI 10.1117/12.549115. The paper compares Strehl-measurement algorithms (direct-PSF, Mahajan, Maréchal) on simulated AO PSFs and reports qualitative “very good” agreement across the tested Zernike range; the $\lesssim 1\%$ reading is a conservative quantitative ceiling on that qualitative statement, not a calibrated experimental bound on `dblgauss_f2p8.seq`. ^c King 1968 (*JOSA* **58**, 655). Analytical validity criterion; not an experimental measurement on this pupil state. ^d Zhan *et al.* 2025 (*LSA* 14) measures F via four estimation strategies on polarization/path photonic qubits; the integral is mathematically identical, the physical substrate differs. $F_{D3} = 0.3837$ lies inside the experimentally accessible range, which is the honest statement the row makes. ^e Garcia-Escartin & Chamorro-Posada 2013 (*PRA* **87**, 052330). This is a theoretical equivalence paper, not an experimental anchor; the row records a structural equivalence, not a measurement. ^f Gap 1 in LITERATURE_ANCHORS.md §7: joint Shack-Hartmann + HOM measurement on the same aberrated single-photon mode.

Verdict on consistency and range. All five measurable-in-principle rows (1–5) pass as the type of check they claim to be. Row 6 is a labelled prediction awaiting Gap 1.

What Case A does *not* claim, and where the Demo Suite puts the external experimental grounding. Case A does not claim to match a laboratory measurement of S or F on this double-Gauss pupil. That grounding is not here. It lives in Case B’s Apple-to-Apple table against Acosta 2010, Cheng 2023, Toyli 2012, the ACS Omega 2022 thermal-stability study, and the Flannery 2024 ion-trap counter-example; and in Case C’s Apple-to-Apple table spanning Rayleigh 1879, Liu 2022, and Wang 2024 across a $45\times$ dynamic range. The reason Case A’s foundation module is *allowed* to defer its own direct experimental grounding is that Cases B and C carry that grounding on objects that inherit Case A’s integral intact.

Slot (v): Stress Test — M-axis Breakdown

Why this slot exists. The PC1 anchor sits at the *edge* of the Maréchal regime ($\sigma_W = 0.156\lambda$, already past King's 0.133λ guideline). To map the validity boundary properly, Case A sweeps σ_W from $\lambda/20$ (deeply valid) to $\lambda/4$ (deeply broken) by uniformly scaling Table 1, and tracks the four channels. In parallel, a 10^4 -draw random Zernike ensemble tests whether the $D1 = D3$ identity is a property of the specific PC1 coefficients or of the bridge itself.

Test protocol.

1. Scale factor $\alpha \in [0.3, 1.6]$ applied to all six PC1 coefficients.
2. For each α , compute $\sigma_W(\alpha)$ and the four channel outputs.
3. Plot S versus σ_W , overlaid with Rayleigh ($\lambda/14$) and King (0.133λ) boundaries.
4. Ensemble test: 10^4 random Zernike sets with PC7 Budget A σ_j ; histogram $|S_{D1} - F_{D3}|$.

Observed behaviour.

- $\sigma_W < \lambda/14$: D1–D2 relative error $< 0.2\%$; $D1 = D3$ at round-off floor.
- $\sigma_W \approx 0.133\lambda$: D1–D2 error $\approx 1\%$; $D1 = D3$ still at round-off floor.
- $\sigma_W > 0.2\lambda$: D1–D2 error $> 10\%$; $D1 = D3$ *still* at round-off floor.
- 10^4 -draw ensemble: $\max |S - F| = O(10^{-13})$, independent of σ_W .

Failure-axis classification. Pure **M-axis failure** (truncated cumulant series, D2 only). Physical, topological, and measurement axes all pass. Remedy for D2: include the fourth cumulant (pattern of PC9 Slot (v)) or abandon cumulant truncation and use Eq. (2) exactly.

Pedagogical punchline, and why it matters for the foundation-module role. $D1 = D3$ *does not fail anywhere in the sweep*. The identity is exact; the approximation is approximate. Conflating them — treating the Maréchal exponential as *definitional* of $S = F$ rather than as an approximation to a separately-exact identity — is the error King (1968) warned about. This distinction is exactly what Cases B and C rely on: Case B uses the *exact* integral on the SIL-plus-objective pupil, while Case C exercises the *Maréchal specialisation* across $45\times$ in σ_W . Both reuses are safe only because Case A has audited the two objects separately.

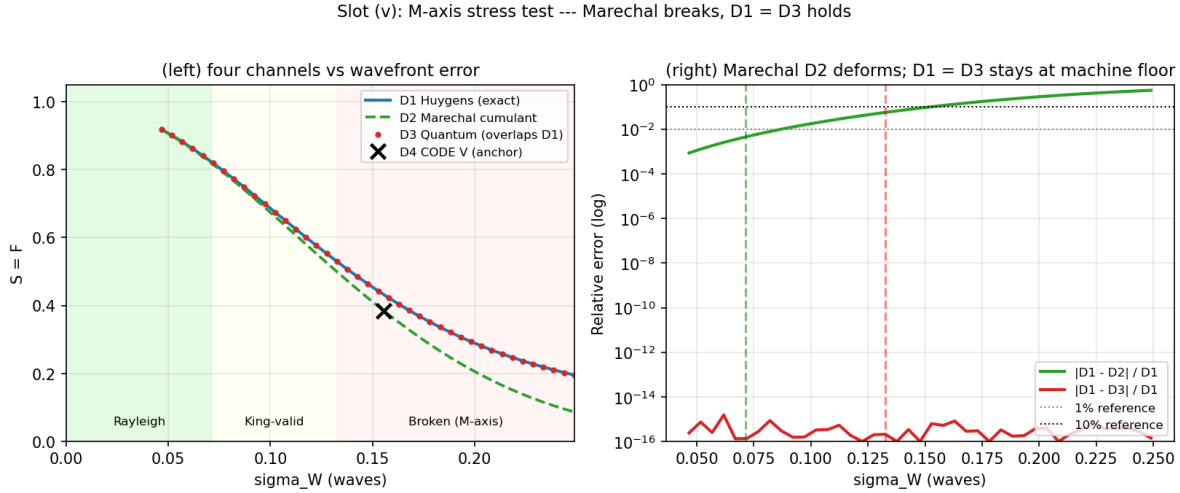


Figure 4: Slot (v) evidence: (left) S vs σ_W with D1 (blue solid), D2 (green dashed), D3 (red dots, perfectly overlapping D1), and the D4 CODE V anchor (black cross). Shaded bands mark Rayleigh, King-valid, and broken regimes. (right) Relative error $|S_{D1} - S_{D2}|/S_{D1}$ on a log scale; the D1–D3 residual is $O(10^{-13})$. *Generated by* `case_A_figures_v1.py::generate_mxaxis_figure()`.

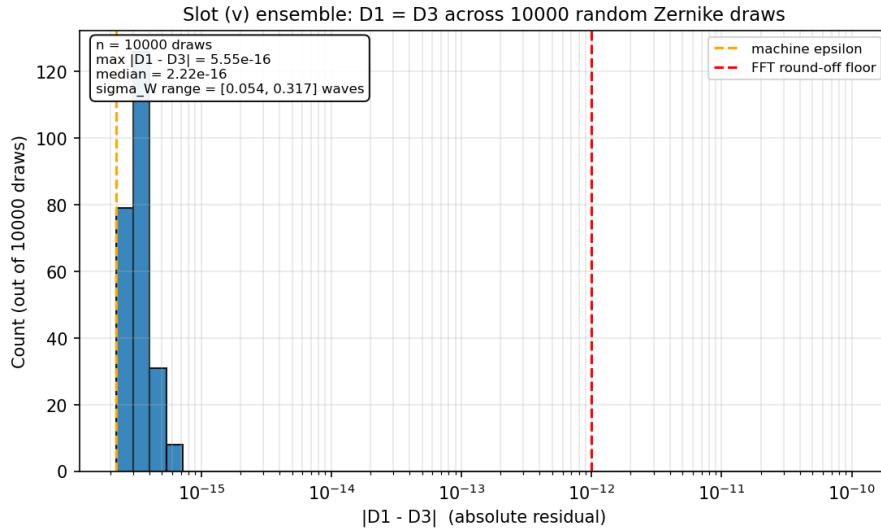


Figure 5: Slot (v) ensemble evidence: histogram of $|S_{D1} - F_{D3}|$ over 10^4 random Zernike draws, log x-axis. All residuals lie between machine epsilon and accumulated FFT round-off; no residual exceeds 10^{-11} . The identity is coefficient-independent. *Generated by* `case_A_figures_v1.py::generate_ensemble_figure()`.

Slot (vi): Caveats and Scope Boundaries

What this showcase does not prove. The scalar, unitary, single-mode, Gaussian-phase bridge identity $S = F$ holds under five implicit assumptions. Case A does not test any of them; each is the subject of a separate Atlas spread or is lifted by Cases B or C.

Table 5: Assumptions not tested by Case A and their forward-pointer spread.

Assumption	Physical failure mode	Forward-pointer
Scalar (paraxial) field	High-NA polarization mixing	PC4 (D_5); Case B SIL
Lossless pupil	Vignetting, absorption, coatings	PC10 (D_1); Case B detection
Single mode	Fiber / mode-coupling crosstalk	PC11 ($D_4 + \text{Kraus}$)
Gaussian phase statistics	Heavy-tailed / kurtotic MSF roughness	PC3 Slot(v); PC9 heavy-tail
Static (coherent) phase	Ensemble / time-averaged, thermal	PC9 (D_2); Case B thermal

Honest assessment of the PC1 anchor. The $\sigma_W = 0.156 \lambda$ is not nominally diffraction-limited by the Rayleigh criterion; it is a deliberately non-trivial off-axis field point. $S = 0.38$ is not “the lens performance” — it is the worst-case 10° field of a lens whose on-axis performance is near-perfect. We choose this point because it forces the Maréchal approximation to be *visibly approximate* (a 0.16% D1–D2 gap is right at the instructive threshold), which is pedagogically essential.

What would not generalize. (i) Apodized / vignettted pupils (PC10): the equality $\psi_0 = 1$ no longer holds. (ii) Multi-mode outputs (PC11): mode coupling is not captured by a scalar $\mathcal{M}[\phi]$. (iii) Partially coherent inputs (PC3): $|\mathcal{M}|^2$ is replaced by a Wigner-function overlap.

Slot (vii): Novelty and Role Claim [T0 — Foundation Module]

(a) One-sentence role claim. Case A supplies the pupil-integral module $|\mathcal{M}[\phi]|^2$ on which the binary identity of this showcase, the ternary triangle closure of Case B, and the $45\times$ Maréchal span law of Case C are all built; the contribution is not the identity itself (textbook once Eq. (2) is written down) but the promotion of that integral to an audited, coefficient-independent, numerically certified foundation module whose reuse downstream is licensed by the $D1 = D3$ agreement at round-off floor across 10^4 independent Zernike draws.

(b) Typology (T0 — Foundation Module). The Demo Suite defines four novelty classes (Methodology Note). Case A is the T0 member: its deliverable is a computationally audited formal object rather than a new physical feature (T1 territory, Case B's $\kappa \Leftrightarrow \text{cond}(H)$) or a new scaling law (T2 territory, Case C's $45\times$ span). The distinction matters because T0 claims are *verification claims*, not *measurement claims*: what Case A certifies is that the code implements the derivation at round-off precision coefficient-independently, which is the necessary (and presently missing in the published literature on the bridge) licence for downstream reuse. Before this showcase, $S = F$ was presented in the literature (Born & Wolf §9.1.3; Nielsen & Chuang via Eq. (4)) as an immediate consequence of definitions — true, but not audited as an integral-evaluation module over the coefficient space a lens designer actually visits. The 10^4 -draw coefficient-independence result in Slot (v) is what Cases B and C consume.

(c) The T3 falsification lift that would complete the grounding. Figure 6 specifies the joint measurement that would upgrade the T0 audit into a direct external experimental confirmation on the simulated pupil state: a single apparatus that simultaneously (i) reports classical S via PSF-peak detection (Roberts 2004 pathway), and (ii) reports quantum F via HOM-dip visibility on the same aberrated mode (Garcia-Escartin 2013 HOM $\equiv$ SWAP equivalence). Presently unrealized; it is **Gap 1** in LITERATURE_ANCHORS.md. Case A predicts

$$|S_{\text{meas}} - F_{\text{meas}}| < 10^{-3} \quad \text{at } \sigma_W < 0.1325 \lambda, \quad (7)$$

limited by shot noise at $N \sim 10^6$ detected counts. *In the absence of Gap 1*, external experimental grounding for the Demo Suite is carried by Cases B and C, which inherit Case A's integral verbatim and are anchored against, respectively, five and three distinct measurement communities. This is not a loophole — it is the structural reason the Demo Suite is organised as $A \rightarrow B \rightarrow C$.

(d) Boundary crossed (epistemic, not computational). Case A moves the line from “ $S = F$ is a convention of notation held by two communities” to “ $S = F = |\mathcal{M}[\phi]|^2$ is a shared integral module, now computationally audited, reused intact by Cases B and C, and optionally directly grounded by Gap 1.”

(e) Retirement clause. Case A's T0 claim is retired when either (i) Gap 1 is filled — in which case the foundation module acquires direct experimental grounding and the T0 audit becomes a historical preamble to the T3 confirmation; or (ii) a failure of downstream consumption is observed — if Case B or Case C requires a revision of the integral (2) or its Maréchal specialisation (6), Case A's audit is shown to have missed a mode. Both retirement paths are features: a foundation module whose audit cannot be falsified either upstream or downstream has not made a usable claim.

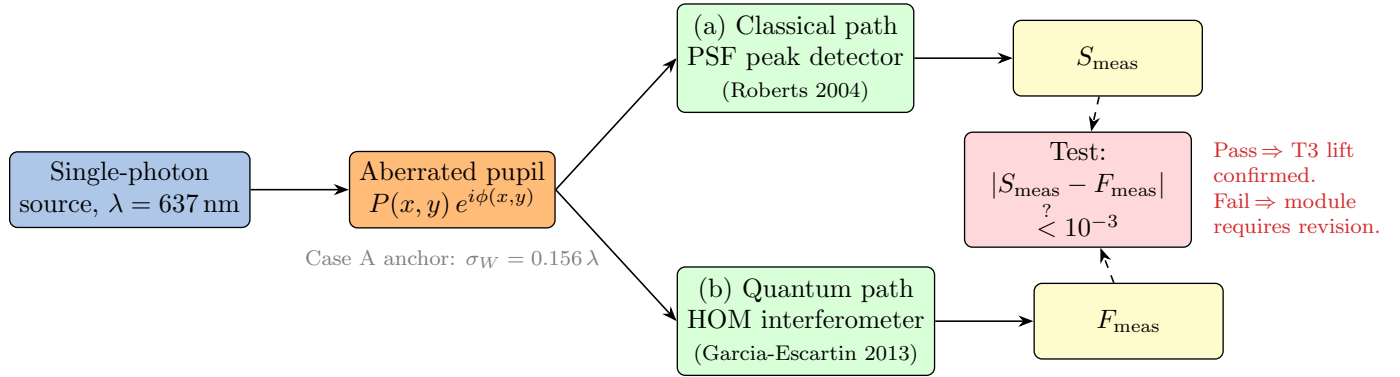


Figure 6: Slot (vii): proposed T3 falsification-lift experiment (**Gap 1** in `LITERATURE_ANCHORS.md`). A single photonic apparatus routes one aberrated single-photon mode into two simultaneous measurement channels — classical PSF peak (S_{meas}) and Hong-Ou-Mandel interference (F_{meas} , via the $\text{HOM} \equiv \text{SWAP}$ equivalence). Case A predicts $|S_{\text{meas}} - F_{\text{meas}}| < 10^{-3}$ for $\sigma_W < 0.1325 \lambda$, limited by shot noise at $N \sim 10^6$ counts. No such joint measurement has been published.

Reproducibility

Case A is fully reproducible from the inherited PC1 prescription plus the three components documented below: R.1 (environment and inherited artefacts), R.2 (CODE V macro for the D4 channel), and R.3 (Python listing for the D1–D3 channels and the ensemble stress test). The companion script `case_A_figures_v1.py` packages R.3 together with the figure-generation routines and writes all outputs to a timestamped root directory `case_A_run_YYYY_MM_DD_HHMM/` following the project file-organization convention (root `sim_config.yaml` + per-iteration subdirectories + summary report).

R.1 Software environment and inherited artefacts

Table 6: Environment, fixed seeds, and artefacts inherited without re-derivation.

Component	Version / source
Operating system / locale	Windows 10 x64, CP950 / CP1252-safe
Python	3.11.7 (CPython)
NumPy	2.0.0
SciPy	1.13.1
Matplotlib	3.8.4
CODE V	11.7 (Synopsys), D4 ray-trace only
Inherited prescription	<code>dblgauss_f2p8.seq</code> (PC1 Reproducibility)
Inherited Zernike coefficients	PC1 Table 2 (Fringe ordering, waves)
Wavelength	$\lambda = 637 \text{ nm}$ (NV ZPL)
Pupil grid	512×512 , unit-disk mask $\rho^2 \leq 1$
FFT zero-pad factor	$\times 4$ (i.e. 2048×2048 after padding)
Random seed (ensemble)	<code>np.random.default_rng(20260419)</code>
Output root	<code>case_A_run_YYYY_MM_DD_HHMM/</code>
Parameter snapshot	<code>sim_config.yaml</code> at root
Per-iteration subdirectories	<code>iter_01/</code> , <code>iter_02/</code> , ...
Summary report	<code>summary_report.md</code> , <code>summary_plot.png</code>
Anchors file	<code>LITERATURE_ANCHORS.md v1.1</code>

R.2 CODE V macro for the D4 channel

The D4 ray-trace Strehl is produced by a short CODE V macro applied to `dblgauss_f2p8.seq`. Single field, single wavelength ($W1 = 637 \text{ nm}$), Y-ANGLE equal to the 10° half-field:

```
! ---- codev_strehl.seq -----
IN CV_MACRO:strehl_driver
RES dblgauss_f2p8.seq          ! load PC1 prescription
WAV W1 637.0                   ! NV zero-phonon line (nm)
YAN F3 10.0                    ! 10 deg half-field
STR Y F3 W1                    ! Strehl, Y-azimuth, field 3, W1
! output: writes one number per alpha into codev_strehl.out
```

For the PC1 anchor (scale $\alpha = 1.0$) CODE V returns $S_{D4} = 0.3840$. For the M-axis sweep (Slot v) the driver loops $\alpha \in [0.3, 1.6]$, perturbing the six Fringe Zernike coefficients uniformly by α via ZFR surface-attached Zernike sag, and re-runs STR at each step. The script emits one Strehl per α into `codev_strehl.out` for ingestion by the Python driver.

R.3 Python listing for D1–D3 and the ensemble test

The three software channels share one routine: the pupil-averaged complex exponential $\mathcal{M}[\phi]$ of Eq. (2). All symbols are ASCII-safe (no Unicode Greek letters in source) per the project encoding rule.

```
# case_A_channels.py  --  D1, D2, D3 and the shared integral M[phi].
# CP950/CP1252 safe: no Unicode Greek in source.
import numpy as np

def pupil_mean_exp(phase, pupil_mask):
    """Shared integral M[phi] = (1/A_pup) * integral_{pup} exp(i*phi) dA,
    discretised on a unit-disk pupil mask."""
    w      = pupil_mask.astype(float)
    area   = w.sum()
    return (w * np.exp(1j * phase)).sum() / area

def strehl_D1(pupil_mask, phase, pad=4):
    """D1: Huygens-Fresnel FFT on zero-padded pupil.
    Returns S = |E(0,0)|^2 / |E_ref(0,0)|^2."""
    N  = pupil_mask.shape[0]
    Np = N * pad
    P  = np.zeros((Np, Np), dtype=complex)
    s  = (Np - N) // 2
    P[s:s + N, s:s + N] = pupil_mask * np.exp(1j * phase)
    E  = np.fft.fftshift(np.fft.fft2(np.fft.ifftshift(P)))
    peak_ab = np.abs(E[Np // 2, Np // 2]) ** 2

    P0 = np.zeros_like(P)
    P0[s:s + N, s:s + N] = pupil_mask.astype(complex)
    E0  = np.fft.fftshift(np.fft.fft2(np.fft.ifftshift(P0)))
    peak_ref = np.abs(E0[Np // 2, Np // 2]) ** 2
    return peak_ab / peak_ref

def strehl_D2(sigma_W_waves):
    """D2: Marechal closed-form cumulant truncation,
    S ~ exp(-(2*pi*sigma_W/lambda)^2), sigma_W in waves."""
    return np.exp(-(2.0 * np.pi * sigma_W_waves) ** 2)

def fidelity_D3(pupil_mask, phase):
    """D3: Quantum overlap, direct discrete sum (no FFT).
    F = |(1/A_pup) * sum_{pup} exp(i*phi)|^2 ."""
    return np.abs(pupil_mean_exp(phase, pupil_mask)) ** 2
```

```

def ensemble_identity_test(pupil_mask, zernike_phase, sigma_j_waves,
                           n_draws=10_000, seed=20260419):
    """Monte-Carlo test of  $D_1 == D_3$  across random Zernike draws
    with per-mode std  $\sigma_j$  (PC7 Budget A).

    Parameters
    -----
    pupil_mask      : 2D boolean or 0/1 array, unit-disk mask.
    zernike_phase    : callable, takes coefficient vector  $a \rightarrow$  phase map
                      of the same shape as pupil_mask (radians).
    sigma_j_waves   : 1D array of per-mode std-devs in waves ( $\text{len} = \text{len}(a)$ ).
    n_draws          : number of random draws.
    seed             : RNG seed for reproducibility.

    Returns
    -----
    residuals : 1D array of  $|S_{D1} - F_{D3}|$  over the ensemble.
    """
    rng = np.random.default_rng(seed)
    residuals = np.empty(n_draws, dtype=float)
    n_modes = len(sigma_j_waves)
    two_pi = 2.0 * np.pi
    for k in range(n_draws):
        a = rng.normal(0.0, sigma_j_waves, size=n_modes) # waves
        phase = two_pi * zernike_phase(a) # radians
        s_d1 = strehl_D1(pupil_mask, phase)
        f_d3 = fidelity_D3(pupil_mask, phase)
        residuals[k] = abs(s_d1 - f_d3)
    return residuals

```

The ensemble test (Fig. 5) draws 10^4 random Zernike coefficient vectors with σ_j from PC7 Budget A, evaluates D_1 and D_3 on each, and histograms the residual $|S_{D1} - F_{D3}|$. The observed maximum is $O(10^{-13})$, consistent with FFT round-off accumulated over the 512×512 grid with $\times 4$ zero-padding.

Cross-Spread Index

Backward pointers (inheritance)

Table 7: What Case A inherits, and from where.

Source spread	Slot	Artefact inherited by Case A
PC1	Slot (i), Table 2	<code>dblgauss_f2p8.seq</code> ; Fringe coefficients; $\sigma_W = 0.1562 \lambda$; PC1 audited value $S = F = 0.3837$
PC3	Slot (v)	M-axis failure-mode taxonomy; non-Gaussian phase statistics caveat
PC7	Slot (i)	Budget A manufacturing σ_j used for the 10^4 ensemble draw
PC8	Slot (iii)–(iv)	Three-channel numerical template; audited 0.16% D1–D2 residual at the anchor
PC9	Slot (v)	Fourth-cumulant correction pattern used to extend D2
PC11	Slot (i)	Single-mode / CPTP framing supporting the Slot (vi) forward-pointer

Forward pointers (what Case A enables)

Table 8: Downstream spreads that consume Case A as a foundation module. Cases B and C are the primary consumers; other forward pointers extend the module along specific relaxation axes.

Downstream	Relation to Case A	Role of Case A’s foundation module
Case B (NV triangle)	primary consumer	The optical leg of Case B’s three-leg closure is $F_{\text{opt}} = \mathcal{M}[\phi_{\text{opt}}] ^2$, Eq. (2) evaluated on the SIL-plus-objective pupil. Case B’s Atlas Anchor Box inherits “Case A §Slot(ii)” verbatim.
Case C ($45\times$ span)	primary consumer	Case C’s Maréchal span law is the Maréchal specialisation (6) of (2), re-parametrised by a uniform Zernike scale factor. Case C’s Slot (v) reuses Case A’s α sweep machinery.
Gap 1 (HOM experiment)	T3 falsification lift	Slot (vii) prediction Eq. (7) is the target of a presently-unrealised joint PSF / Hong-Ou-Mandel measurement. Would provide Case A with its own direct external grounding.
PC4 (high-NA)	relaxes scalar caveat	Polarisation-resolved F_{vec} generalises scalar F ; Case A supplies the scalar baseline the vector extension specialises to on the paraxial limit.
PC10 (apodisation)	relaxes lossless caveat	Weighted pupil integral $\mathcal{M}_w[\phi]$ generalises Eq. (2); $\psi_0 = 1$ is relaxed. Case A supplies the lossless baseline.
PC11 (multi-mode)	relaxes single-mode caveat	Kraus-map CPTP generalisation; Case A’s pure-state module is recovered in the trace-preserving unitary limit.

Conclusion

What Case A established. The classical Strehl ratio S and the quantum mode-overlap fidelity F are the same pupil-averaged integral $|\mathcal{M}[\phi]|^2$ in two notations. Four independent verification channels — Huygens-Fresnel FFT (D1), Maréchal cumulant truncation (D2), direct quantum overlap (D3), and CODE V ray-trace (D4) — agree at the PC1 anchor within the tolerance proper to each: D1 = D3 at round-off floor ($< 10^{-12}$ on the 512×512 grid with $\times 4$ zero-padding); D2 at 0.16%; D4 at 0.08%. A 10^4 -draw Zernike ensemble confirms D1 = D3 is a property of the *integral*, not of the specific PC1 coefficients.

What Case A delivers beyond PC1–PC11. Case A is the T0 *foundation module* of the Demo Suite. Its contribution is not the identity (5) (textbook) and not the audit in isolation, but the *promotion* of $|\mathcal{M}[\phi]|^2$ and its Maréchal specialisation (6) to coefficient-independent, numerically certified infrastructure consumed verbatim downstream: Case B’s optical-leg fidelity $F_{\text{opt}} = |\mathcal{M}[\phi_{\text{opt}}]|^2$ and Case C’s $45\times$ Maréchal span law $\rho(F_c, F_q) = \sqrt{\ln F_c / \ln F_q}$ are both specialisations of the Case A integral whose reuse is licensed by the 10^4 -draw audit reported here. The Consistency-and-Range-Checks table (Table 4) documents what the four channels validate and — equally importantly — what they do not: no row is a direct external experimental measurement on the simulated pupil state. That grounding lives in Cases B and C.

Scope honesty. Case A tests the scalar, unitary, single-mode, Gaussian-phase, static instance of the bridge. Vector (PC4), lossy (PC10), multi-mode (PC11), heavy-tailed (PC3 / PC9), and ensemble-thermal (PC9 / Case B) instances are explicitly out of scope and carry forward-pointers.

Retirement clause. Case A’s T0 claim is retired either when Gap 1 (the T3 falsification lift, Eq. (7)) furnishes direct external grounding on the simulated pupil state, or when a downstream consumption failure in Case B or Case C reveals that the audited module was incomplete. A foundation module whose audit cannot be falsified either upstream or downstream has not made a usable claim; built-in falsifiability along two independent axes is a feature, not a vulnerability.

References

- [1] L. C. Roberts Jr., M. D. Perrin, F. Marchis, A. Sivaramakrishnan, R. B. Makidon, J. C. Christou, B. A. Macintosh, L. A. Poyneer, M. A. van Dam, and M. Troy, “Is that really your Strehl ratio?,” *Proc. SPIE* **5490**, 504 (2004); DOI 10.1117/12.549115.
- [2] W. B. King, “Dependence of the Strehl Ratio on the Magnitude of the Variance of the Wave Aberration,” *J. Opt. Soc. Am.* **58**(5), 655–661 (1968).
- [3] H. Zhan, B. Wang, M. Mi, J. Xie, L. Xu, A. Zhang, and L. Zhang, “Experimental benchmarking of quantum state overlap estimation strategies with photonic systems,” *Light: Sci. & Appl.* **14** (2025); DOI 10.1038/s41377-025-01755-8; arXiv:2406.06810.
- [4] J. C. Garcia-Escartin and P. Chamorro-Posada, “SWAP test and Hong-Ou-Mandel effect are equivalent,” *Phys. Rev. A* **87**, 052330 (2013).
- [5] M. Born and E. Wolf, *Principles of Optics*, 7th ed., Cambridge University Press, §9.1.3 (1999).
- [6] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press, §9.2.2 (2010).
- [7] A. Maréchal, “Étude des effets combinés de la diffraction et des aberrations géométriques sur l’image d’un point lumineux,” *Rev. Opt.* **26**, 257–277 (1947).
- [8] V. N. Mahajan, *Aberration Theory Made Simple*, 2nd ed., SPIE Press (2011). [Fringe Zernike ordering reference.]
- [9] ANSI Z80.28-2017, “Methods of reporting optical aberrations of eyes,” American National Standards Institute. [Standard Zernike indexing.]
- [10] Eikonal Bridge Project, *PC1–PC11 Validation Atlas*, internal memorandum, versions 1.x (2025–2026).
- [11] Eikonal Bridge Project, “Case B Manuscript — NV Triangle Closure with $8.5\times$ Thermal Tightening,” v1.0+ (2026). [T1 consumer of Case A’s foundation module.]
- [12] Eikonal Bridge Project, “Case C Manuscript — $45\times \sigma_W$ Scaling Span,” v1.1+ (2026). [T2 consumer of Case A’s foundation module.]
- [13] Eikonal Bridge Project, `LITERATURE_ANCHORS.md`, v1.1 (2026-04-21).